



Definition (RIA – Robot institute of america)

“ A **reprogrammable, multifunctional** manipulator designed to move material, parts, tools, or specialized devices through various programmed motions for the performance of a variety of tasks”

10/3/2025

3

3



Definition – (International Organization for Standardization -ISO):

“An automatically controlled, reprogrammable, multipurpose manipulator programmable in three or more axes, which can be either fixed in place or mobile for use in industrial automation applications”

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4

4



To qualify as a robot, a machine must be able to:

- 1) Sensing and perception: get information from its surroundings
- 2) Carry out different tasks: Locomotion or manipulation, do something physical—such as move or manipulate objects
- 3) Re-programmable: can do different things
- 4) Function autonomously and/or interact with human beings

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5

5



Three Laws of Robotics

FIRST LAW: Do not harm human being

A robot must not harm a human being, or, through inaction, allow a human being to come to harm.

SECOND LAW: Obey human being

A robot must always obey the orders given it by human beings except where such orders would conflict with the First Law.

THIRD LAW: Protects itself from harm

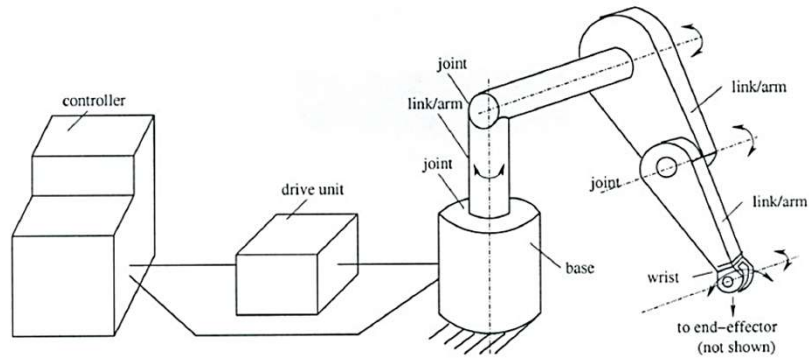
A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

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6

6

Components of robotic system



Various Components : 1.Base 2.Links and Joints 3.End-effector / gripper 4.Wrist 5.Drive / Actuator 6.Controller 7.Sensors

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Fundamentals of robotics by dk pratihar

7

7

Serial Manipulator



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8

8

Parallel Manipulator



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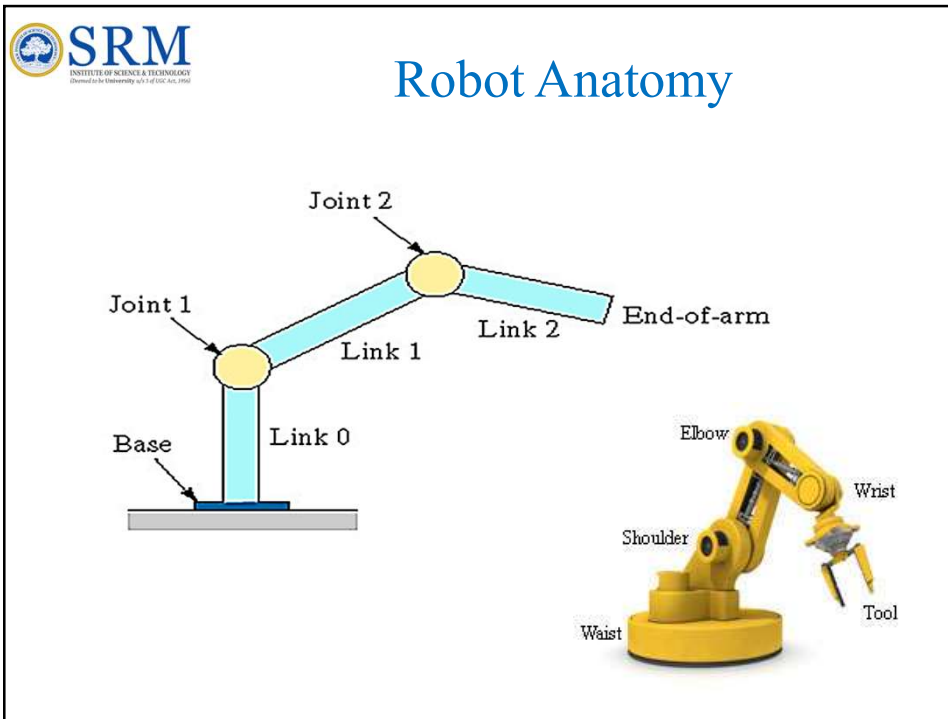
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9

Robot Anatomy

- Manipulator consists of joints and links
 - Joints provide relative motion
 - Links are rigid members between joints
 - Various joint types: linear and rotary
 - Each joint provides a “degree-of-freedom”
 - Most robots possess five or six degrees-of-freedom
- Robot manipulator consists of two sections:
 - Body-and-arm – for positioning of objects in the robot's work volume
 - Wrist assembly – for orientation of objects

10



11



Degrees of freedom (DOF)

- **DOF / Connectivity** is defined as one of the variable required to define the motion of the body in space.
- Degree of freedom of a system depends on the number of variable or coordinates that are needed to describe its position.
- Each joint in a robotic system gives the robot one degree of freedom.

12



The Robotic Joints

- A robot joint is a mechanism that permits relative movement between parts of a robot arm.
- The joints of a robot are designed to enable the robot to move its end - effector along a path from one position to another as desired.

13



The Robotic Joints

- These degrees of freedom, independently or in combination with others, define the complete motion of the end-effector.
- These motions are accomplished by movements of individual joints of the robot arm. The joint movements are basically the same as relative motion of adjoining links.
- Depending on the nature of this relative motion, the joints are classified as *prismatic* or *revolute*.

14



The Robotic Joints

- *Prismatic joints (L)* are also known as sliding as well as linear joints.
- They are called *prismatic* because the cross section of the joint is considered as a generalized prism. They permit links to move in a linear relationship.

15

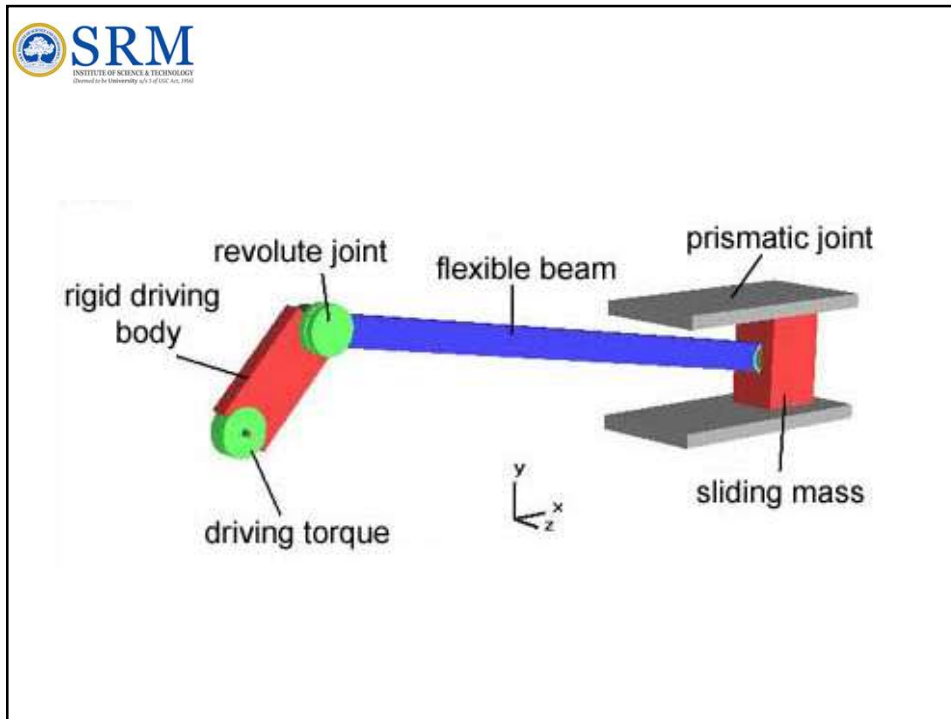


The Robotic Joints

Revolute joints permit only angular motion between links. Their variations include:

- *Rotational joint (R)*
- *Twisting joint (T)*
- *Revolving joint (V)*

16



17



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Mechanical Joints for Robots

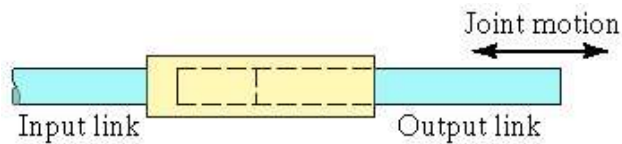
- Translational motion - ***Prismatic joints***
 - Linear joint (type L)
 - Orthogonal joint (type O)
- Rotary motion - ***Revolute joints***
 - Rotational joint (type R)
 - Twisting joint (type T)
 - Revolving joint (type V)

18

Translational Motion Joints

- **Linear joint**

Type L joint; the relative movement between the input link and the output link is a translational sliding motion, with the axes of the two links parallel.

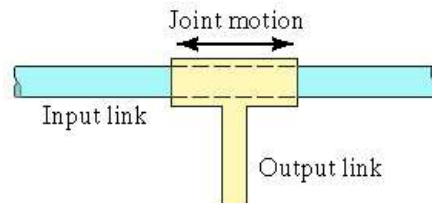


19

Translational Motion Joints

- **Orthogonal joint**

Type O joint; the relative movement between the input link and the output link is a translational sliding motion, but the output link is perpendicular to the input link.

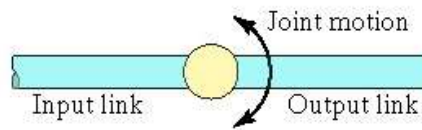


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Rotary motion

- **Rotational joint**

Type R joint; this provides rotational relative motion, with the axis of rotation perpendicular to the axes of the input and output links.

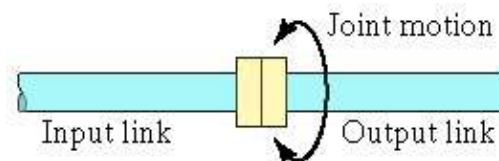


21

Rotary motion

- **Twisting joint**

Type T joint; this provides rotary motion, but the axis of rotation is parallel to the axes of the two links.

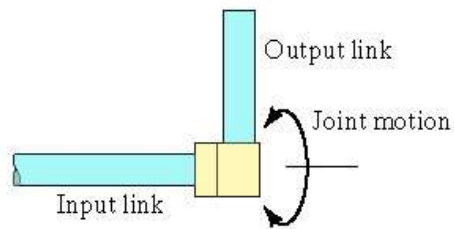


22

Rotary motion

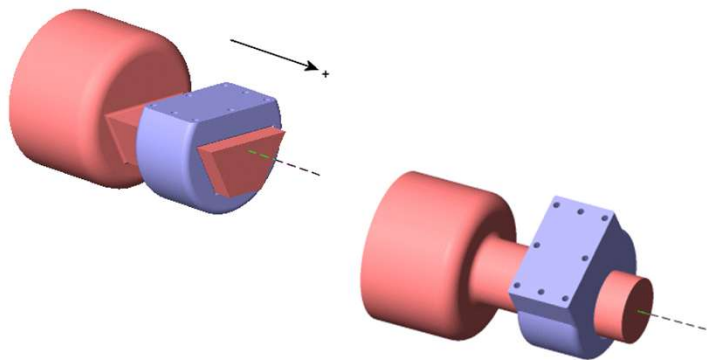
- **Revolving joint**

Type V joint; the axis of the input link is parallel to the axis of rotation of the joint, and the axis of the output link is perpendicular to the axis of rotation.

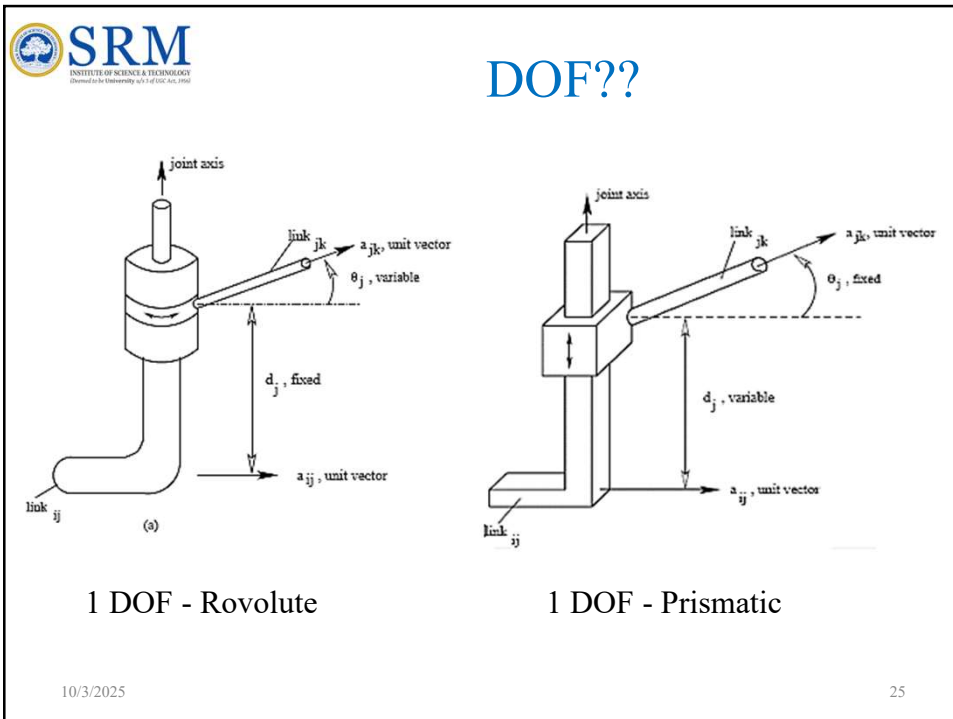


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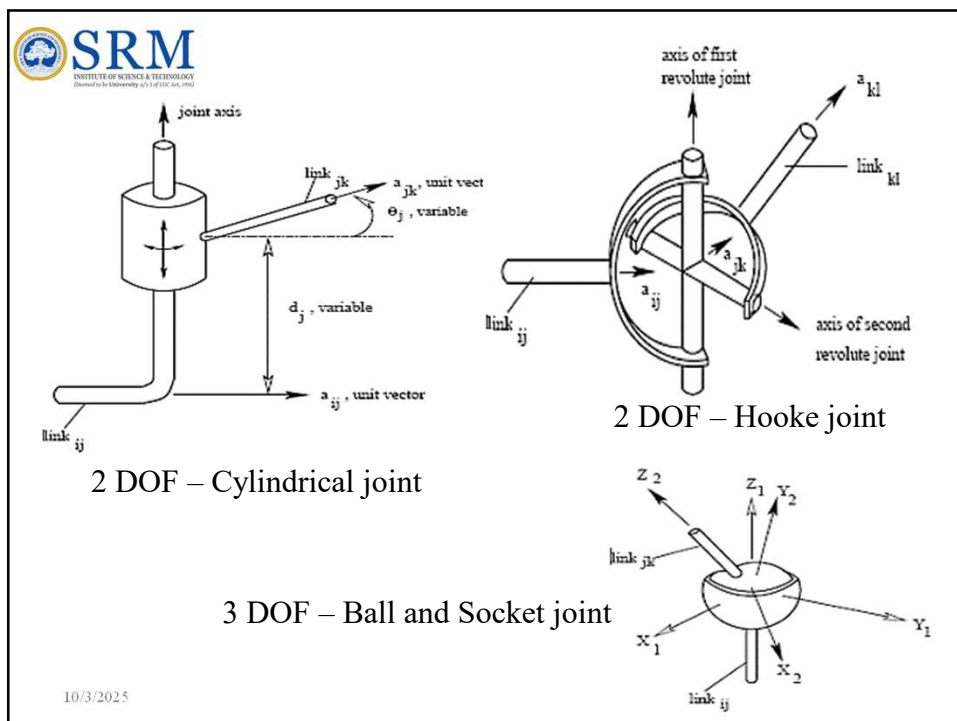
Constraints???



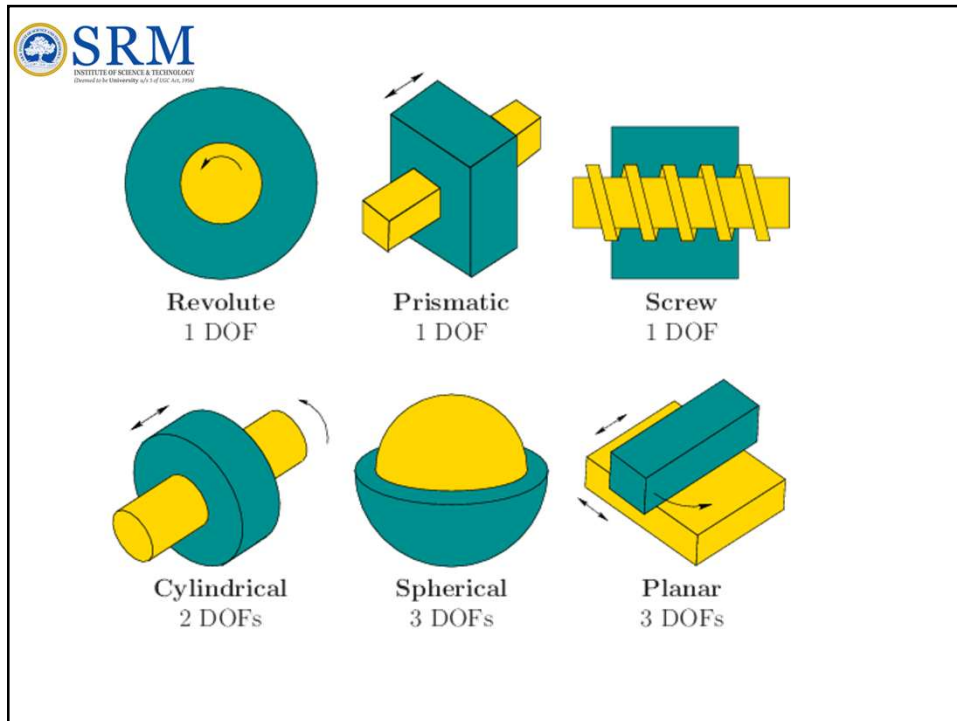
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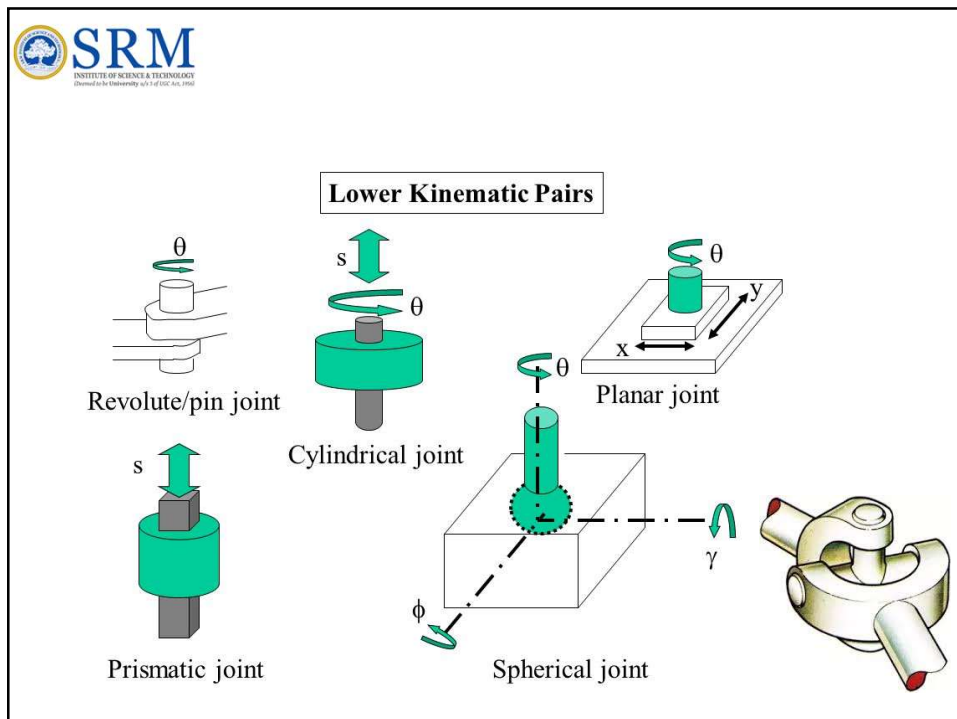
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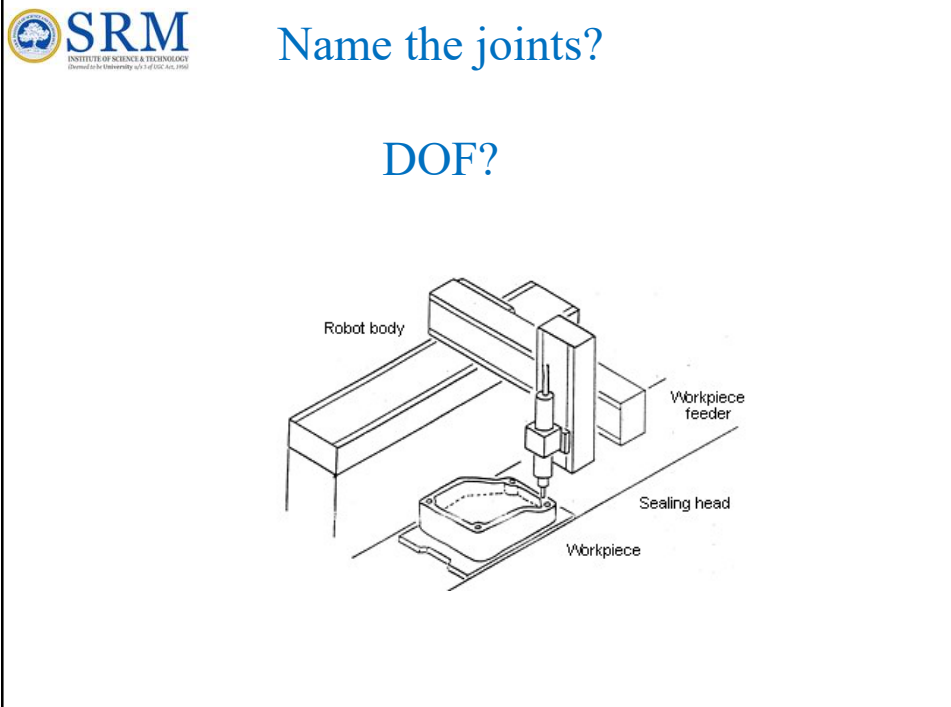
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27



28



29



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DOF

- A point in 2-D: 2 DOF
- A point in 3-D space: 3 DOF
- A rigid body in 3-D: 6 DOF
- Spatial Manipulator: 6 DOF
- Planar Manipulator: 3 DOF

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30

30



Redundant Manipulator:

- Either a Spatial Manipulator with more than 6 dof or a Planar Manipulator with more than 3 dof

Under-actuated Manipulator

- Either a Spatial Manipulator with less than 6 dof or a Planar Manipulator with less than 3 dof

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31

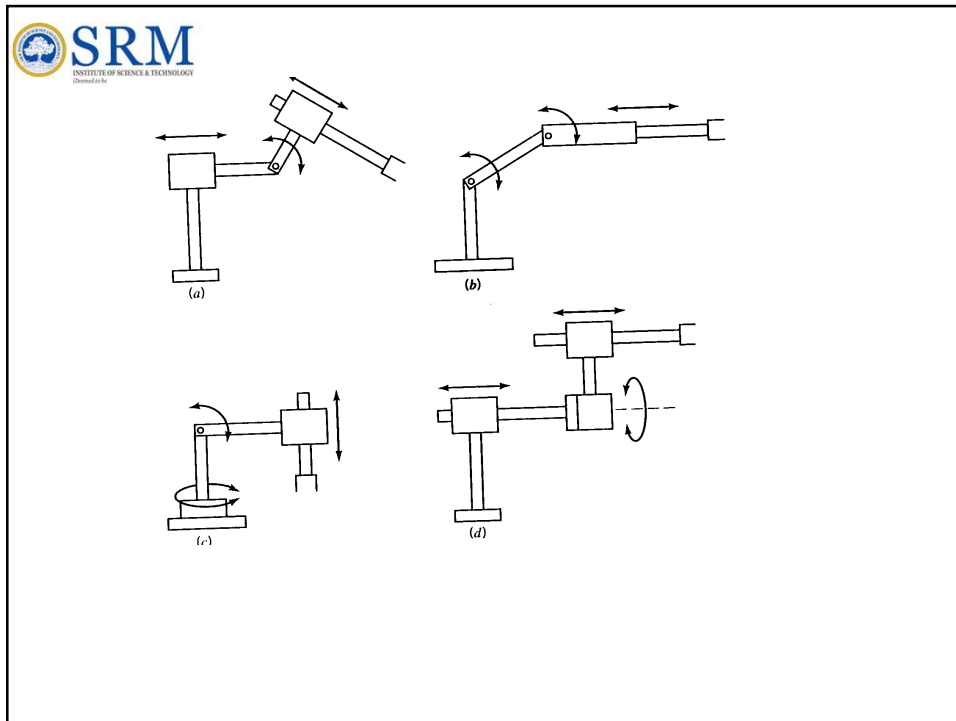
31



Exercise

- Draw the following configurations
 - LRL - ORO
 - RRL - RRO
 - TRL
 - LVL - OVO

32



33

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- Sketch following manipulator configurations
- (a) TRT:R, (b) TVR:TR, (c) RR:T.

Solution:

(a) TRT:R

(b) TVR:TR

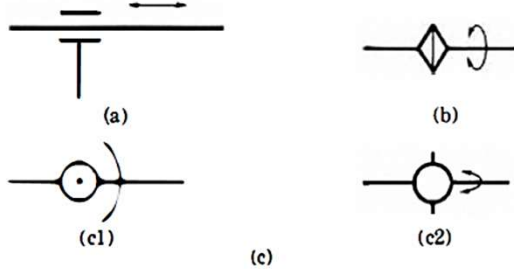
(c) RR:T

34



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Joint representations



(a) (b)

(c1) (c2)

(c)

Symbols of joints (arrows show direction of motion). (a) Prismatic joint. (b) Revolute joint

1. (c) Revolute joint 2. (c1) Up-and-down rotation. (c2) Back-and-forth rotation.

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Foundations of Robotics
Analysis and Control
Tsuneo Yoshikawa

35

35



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Can you name the common coordinate systems in robots????

- Cartesian coordinate system
- Cylindrical coordinate system
- Polar or Spherical coordinate system
- Revolute coordinate system

36



Robot Body-and-Arm Configurations

- Five common body-and-arm configurations for industrial robots:
 1. Polar coordinate body-and-arm assembly
 2. Cylindrical body-and-arm assembly
 3. Cartesian coordinate body-and-arm assembly
 4. Jointed-arm body-and-arm assembly
 5. Selective Compliance Assembly Robot Arm (SCARA)
- Function of body-and-arm assembly is to position an end effector (e.g., gripper, tool) in space

37



Work volume (Work envelope)

- Volume in which the robot is able to work.
- The work volume is determined by the following physical characteristics of the robot:
 - The robots physical configuration
 - The size of the body, arm, and wrist components
 - The limits of the robots joint movements.

Dextrous Workspace

It is the volume of space, which the robot's end-effector can reach with various orientations

Reachable Workspace

It is the volume of space that the end-effector can reach with one orientation

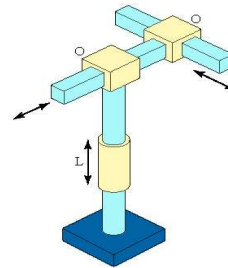
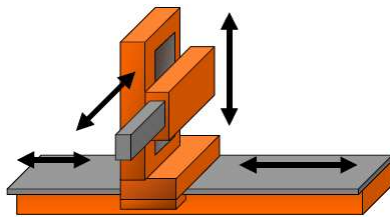
38



ROBOT CLASSIFICATION

Cartesian Configuration:

- Robots with Cartesian configurations consists of links connected by linear joints (L). Gantry robots are Cartesian robots (LLL).
- Also known as rectilinear robot and x-y-z robot. Consists of three sliding joints, two of which orthogonal (O joint).



39

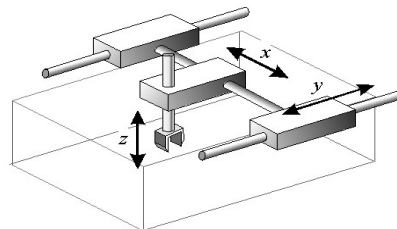


Cartesian Coordinate Body-and-Arm Assembly

A robot with 3 prismatic joints – the axes consistent with a Cartesian coordinate system.

Commonly used for:

- pick and place work
- assembly operations
- handling machine tools
- arc welding



40



➤ Advantages

- Linear motion in three dimension
- Simple kinematic model
- Rigid structure
- Higher repeatability and accuracy
- High lift-carrying capacity as it doesn't vary at different locations in work volume
- Easily visualize
- Can increase work volume easily
- Inexpensive pneumatic drive can be used for P&P operation

41



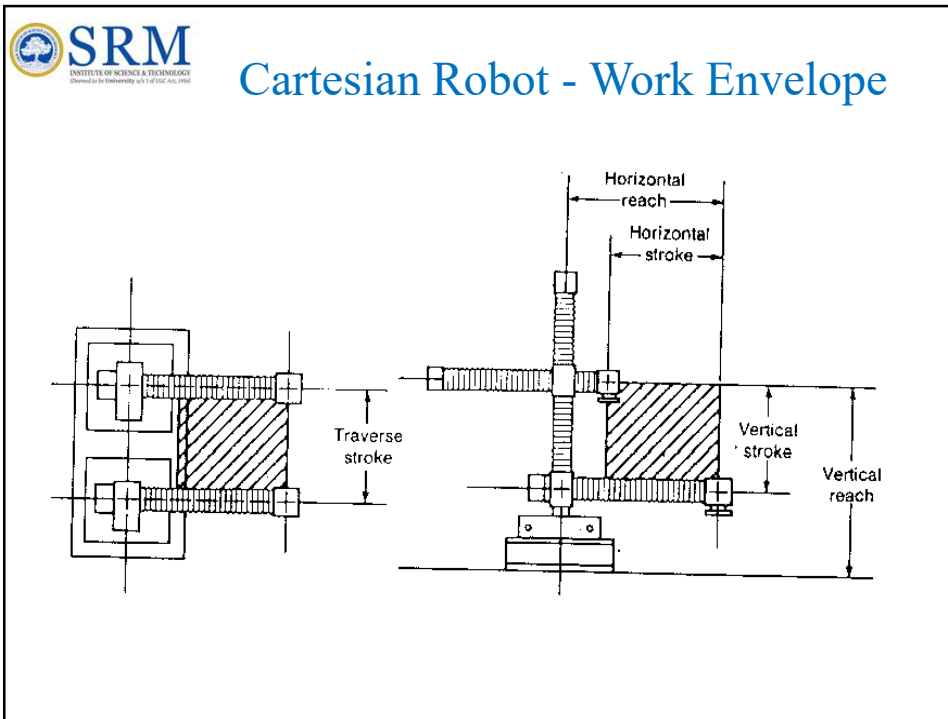
➤ Disadvantages

- requires a large volume to operate in
- work space is smaller than robot volume
- unable to reach areas under objects
- must be covered from dust

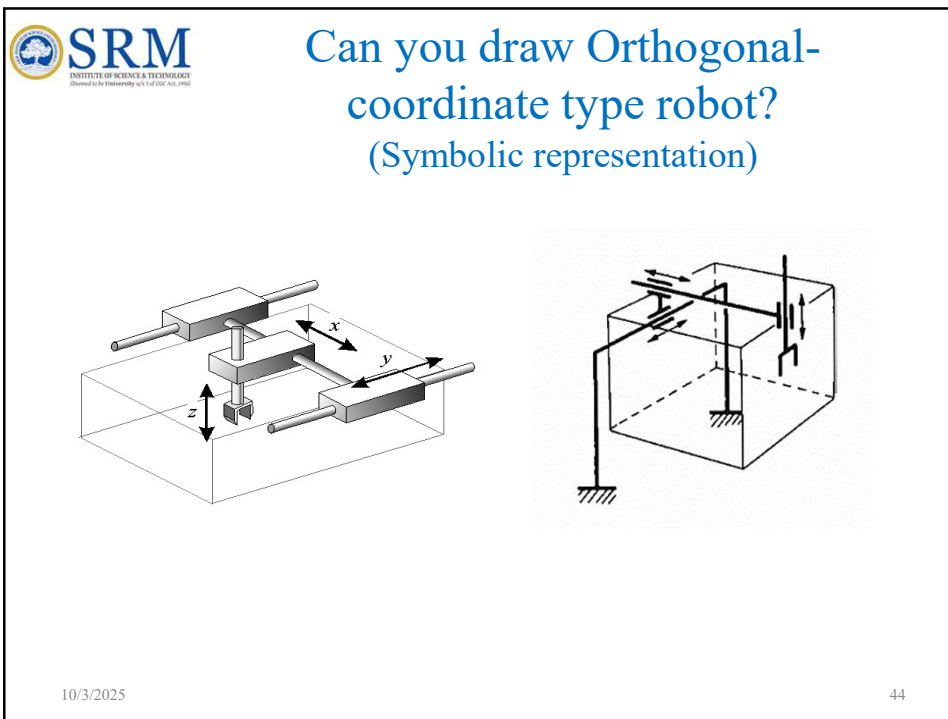
➤ Applications

- Assembly
- Palletizing and loading-unloading machine tools,
- Handling
- Welding

42



43



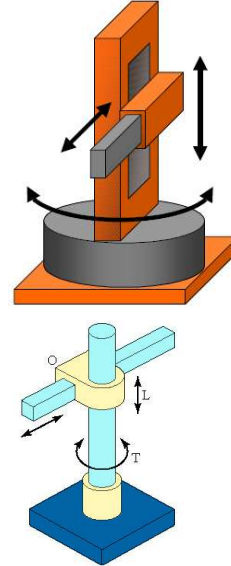
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44

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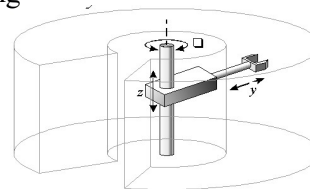
Cylindrical coordinate

Consists of a vertical column, relative to which an arm assembly is moved up or down. The arm can be moved in and out relative to the axis of the column. Common configuration is to use a T joint to rotate the column about its axis. An L joint is used to move the arm assembly vertically along the column, while an O joint is used to achieve radial movement of the arm.

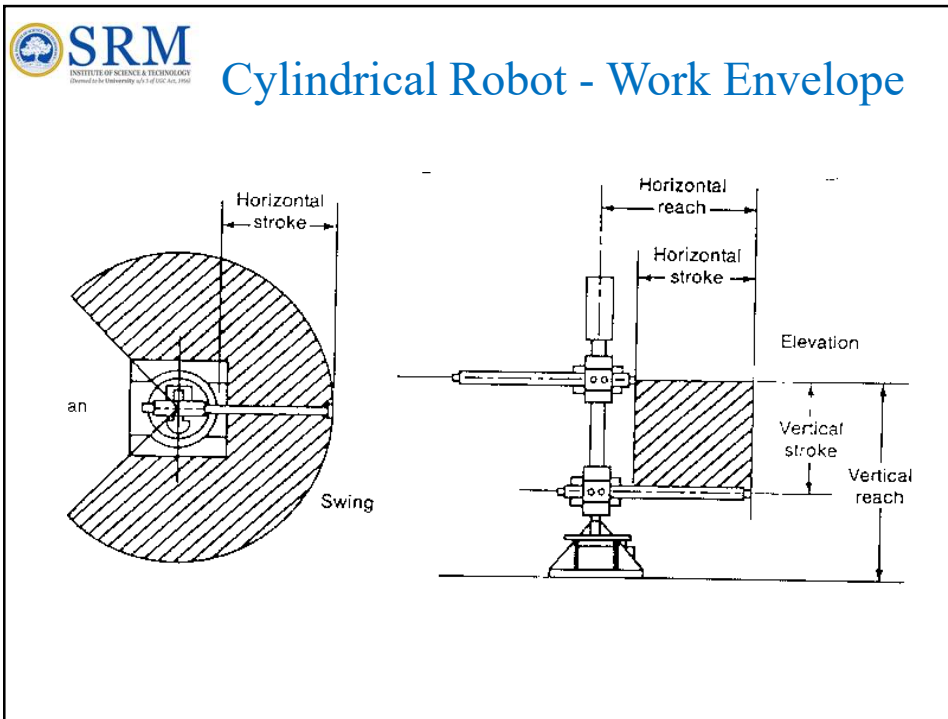


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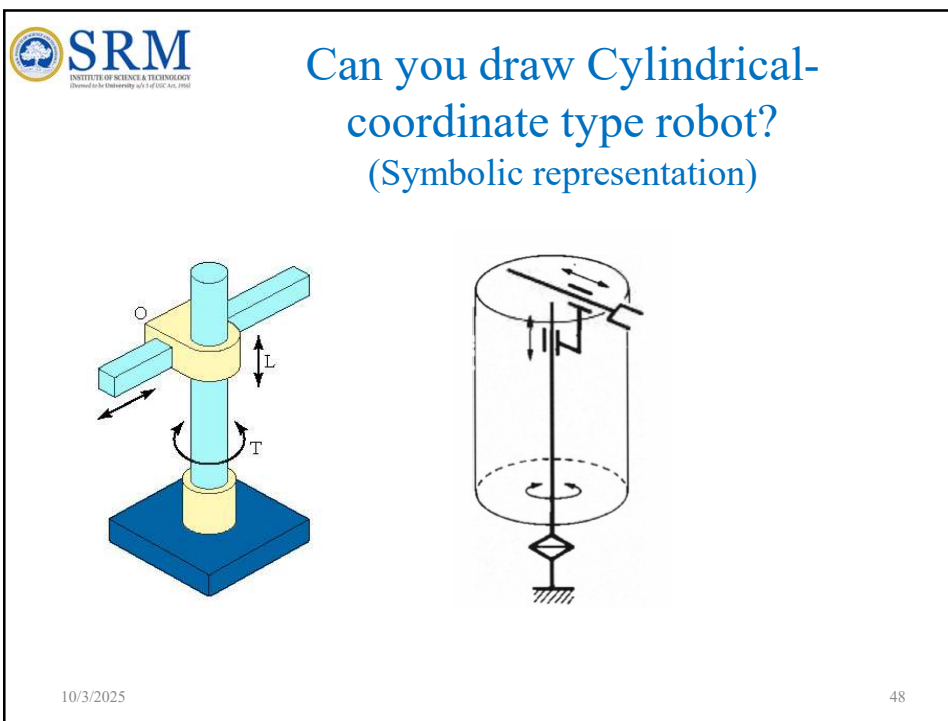
- Advantages
 - Simple kinematic model
 - Rigid structure & high lift-carrying capacity
 - Easily visualize
 - Very powerful when hydraulic drives used
- Disadvantages
 - Restricted work space
 - Lower repeatability and accuracy
 - Require more sophisticated control
- Applications
 - Palletizing, Loading and unloading
 - Material transfer, foundry and forging



46



47

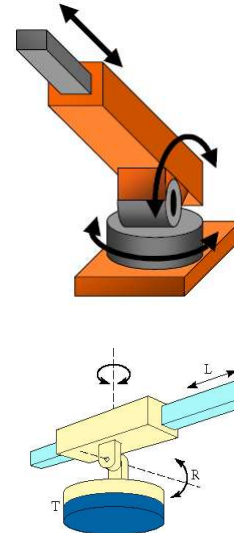


48

48

Polar Coordinate Body-and-Arm Assembly

- Notation TRL:
- Consists of a sliding arm (L joint) actuated relative to the body, which can rotate about both a vertical axis (T joint) and horizontal axis (R joint)
- Polar robots have a work space of **spherical shape**. Generally, the arm is connected to the base with a twisting (T) joint and rotatory (R) and linear (L) joints follow.



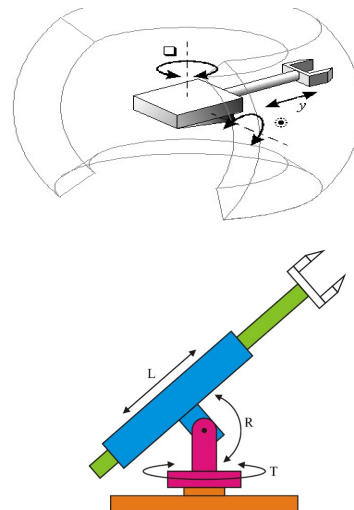
49

Spherical/Polar Robots

A robot with 1 prismatic joint and 2 rotary joints – the axes consistent with a polar coordinate system.

Commonly used for:

- handling at die casting or fettling machines
- handling machine tools
- arc/spot welding



50

➤ Advantages

- Covers a large volume
- Can bend down to pick objects up off the floor
- Higher reach ability

➤ Disadvantages

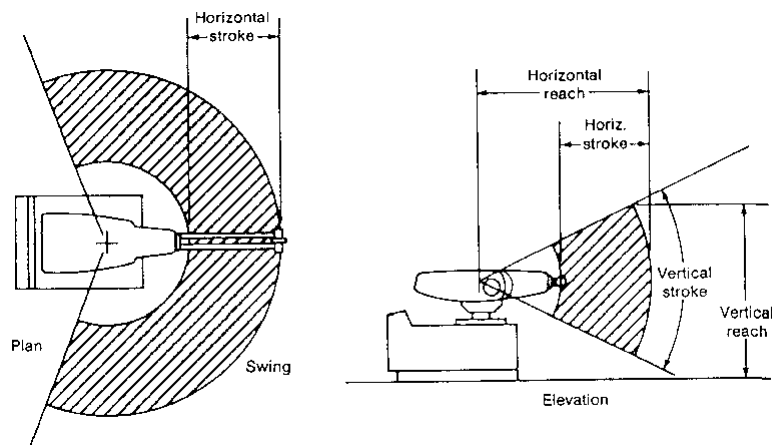
- Complex kinematic model
- Difficult to visualize

➤ Applications

- Palletizing
- Handling of heavy loads e.g. casting, forging

51

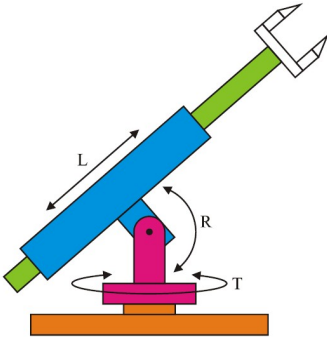
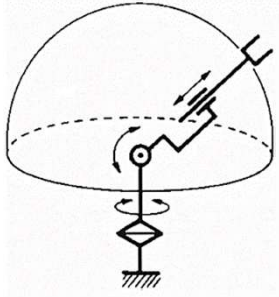
Spherical Robot - Work Envelope



52

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Can you draw Polar-coordinate type robot? (Symbolic representation)

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53

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ROBOT CLASSIFICATION

- The designation of the arm for this configuration can be TRL or TRR.
- Robots with the designation TRL are also called **spherical robots**. Those with the designation TRR are also called **articulated robots**. An articulated robot more closely resembles the human arm.

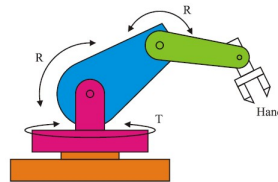
54

Articulated Robots

A robot with at least 3 rotary joints.

Commonly used for:

- assembly operations
- welding
- weld sealing
- spray painting
- handling at die casting or fettling machines



55

Articulated Robots

Advantages:

- all rotary joints allows for maximum flexibility
- any point in total volume can be reached.
- all joints can be sealed from the environment.

Disadvantages:

- extremely difficult to visualize, control, and program.
- restricted volume coverage.
- low accuracy

56

Joint-arm Configuration

- The jointed-arm is a combination of cylindrical and articulated configurations. The arm of the robot is connected to the base with a twisting joint. The links in the arm are connected by rotatory joints. Many commercially available robots have this configuration.
- General configuration of a human arm, this consists of a vertical column that swivels about the base using a T joint. At the top of the column is a shoulder joint (an R joint), output to an elbow joint (another R joint).

57

Joint-arm Configuration



58

➤ Advantages

- Maximum flexibility
- Cover large space relative to work volume objects up off the floor
- Suits electric motors
- Higher reach ability

➤ Disadvantages

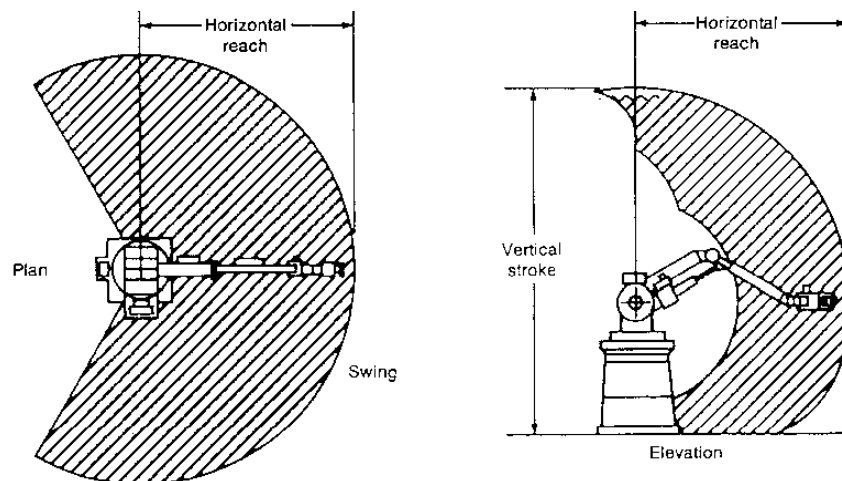
- Complex kinematic model
- Difficult to visualize
- Structure not rigid at full reach

➤ Applications

- Spot welding, Arc welding

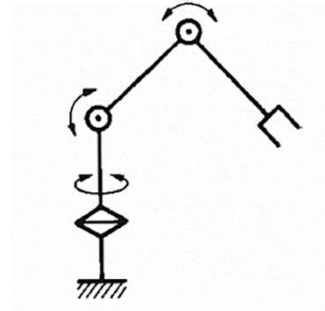
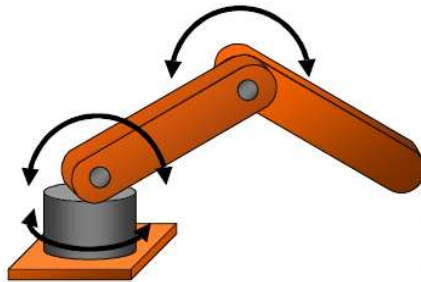
59

Jointed Arm - Work envelope



60

Can you draw Jointed Arm type robot? (Symbolic representation)



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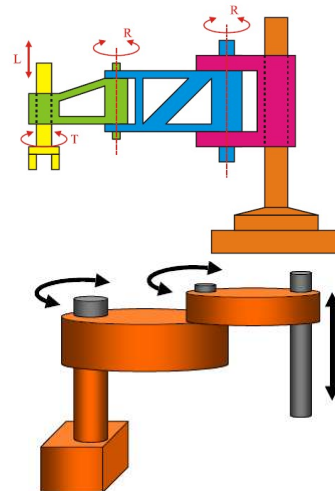
61

61

SCARA Robots

(*Selective Compliance Articulated Robot Arm*)

Selective Compliance Assembly Robot Arm. Similar in construction to the jointer-arm robot, except that the shoulder and elbow rotational axes are vertical, which means that the arm is very rigid in the vertical direction, but compliant in the horizontal direction.



62

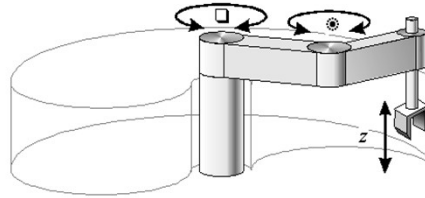
SCARA Robots

(Selective Compliance Articulated Robot Arm)

A robot with at least 2 parallel rotary joints.

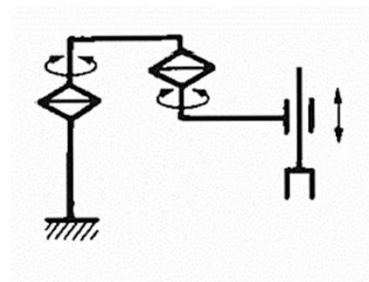
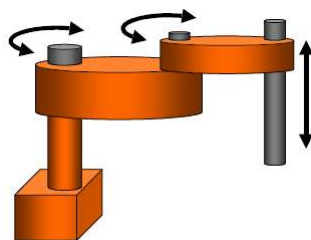
Commonly used for:

- pick and place work
- assembly operations



63

Can you draw SCARA type robot?
 (Symbolic representation)



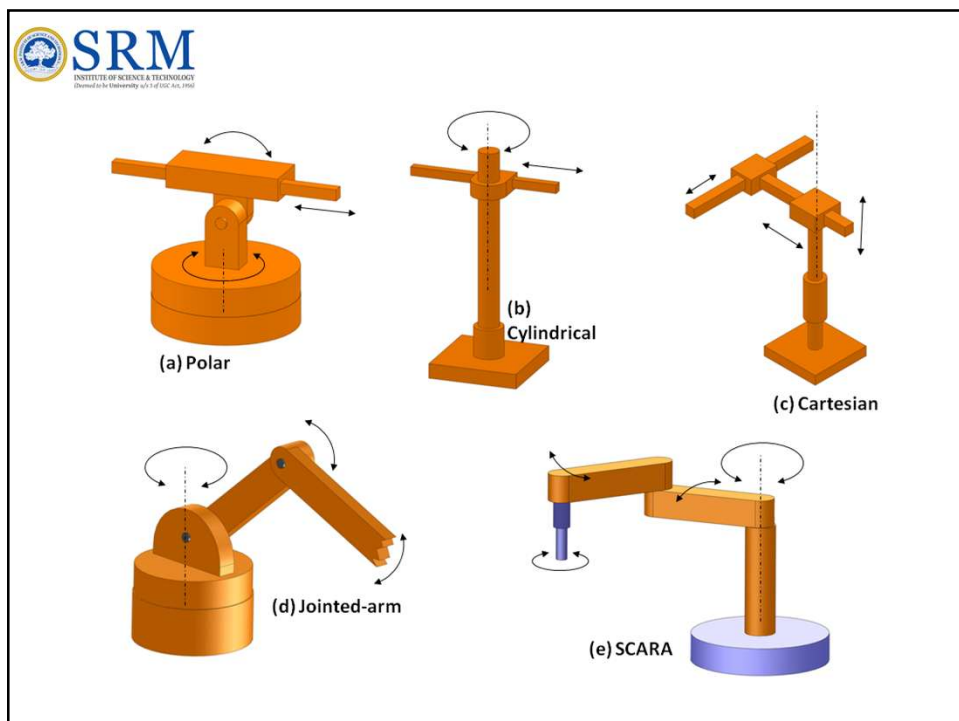
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64

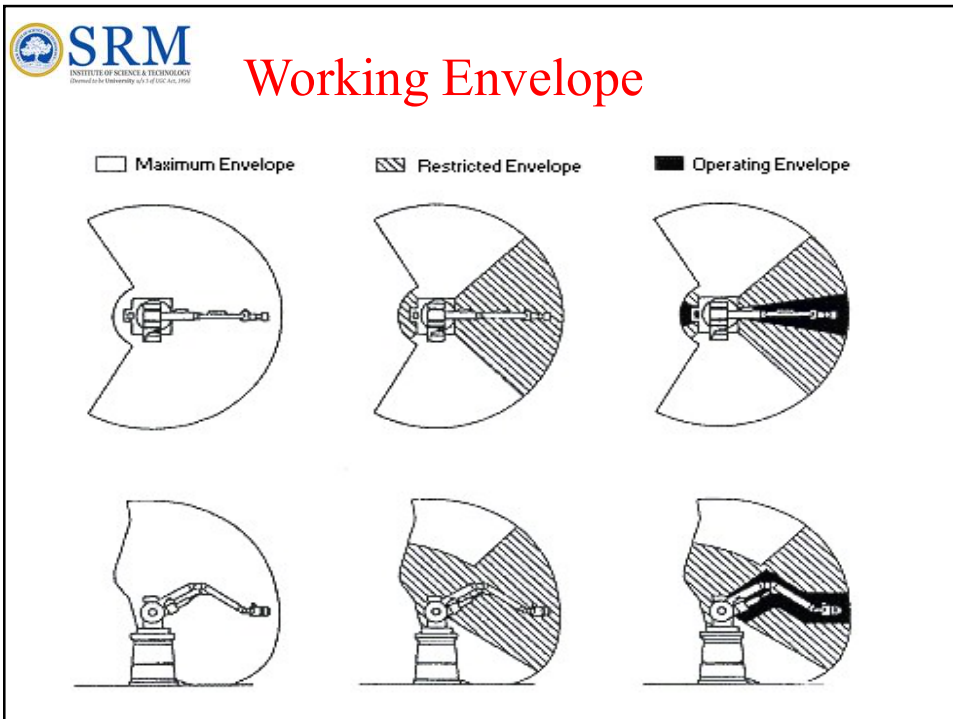
64

 SRM INSTITUTE OF SCIENCE & TECHNOLOGY Deemed to be University act of 1986, Sec. 3	Configuration	Advantages	Disadvantages
	Cartesian coordinates	3 linear axes, easy to visualize, rigid structure, easy to program	Can only reach front of itself, requires large floor space, axes hard to seal
	Cylindrical coordinates	2 linear axes +1 rotating, can reach all around itself, reach and height axes rigid, rotational axis easy to seal	Can't reach above itself, base rotation axis as less rigid, linear axes is hard to seal, won't reach around obstacles
	SCARA coordinates	1 linear + 2 rotating axes, height axis is rigid, large work area for floor space	2 ways to reach point, difficult to program off-line, highly complex arm
	Spherical coordinates	1 linear + 2 rotating axes, long horizontal reach	Can't reach around obstacles, short vertical reach
	Revolute coordinates	3 rotating axes can reach above or below obstacles, largest work area for least floor space	Difficult to program off-line, 2 or 4 ways to reach a point, most complex manipulator

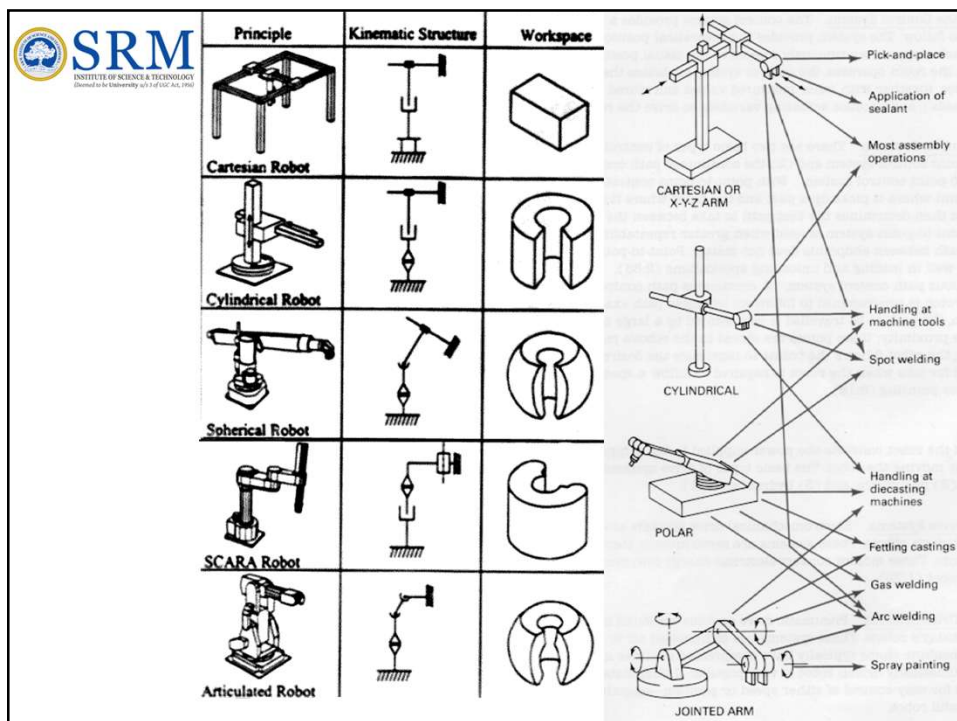
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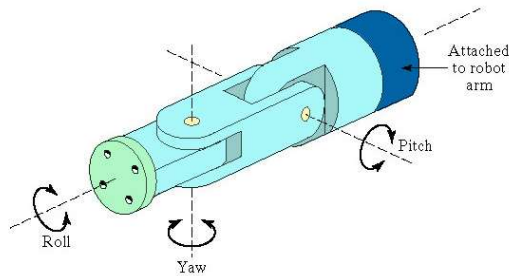
67



68

Wrist Configurations

- Wrist assembly is attached to end-of-arm
- End effector is attached to wrist assembly
- Function of wrist assembly is to orient end effector
 - Body-and-arm determines global position of end effector
- Two or three degrees of freedom:
 - Roll
 - Pitch
 - Yaw

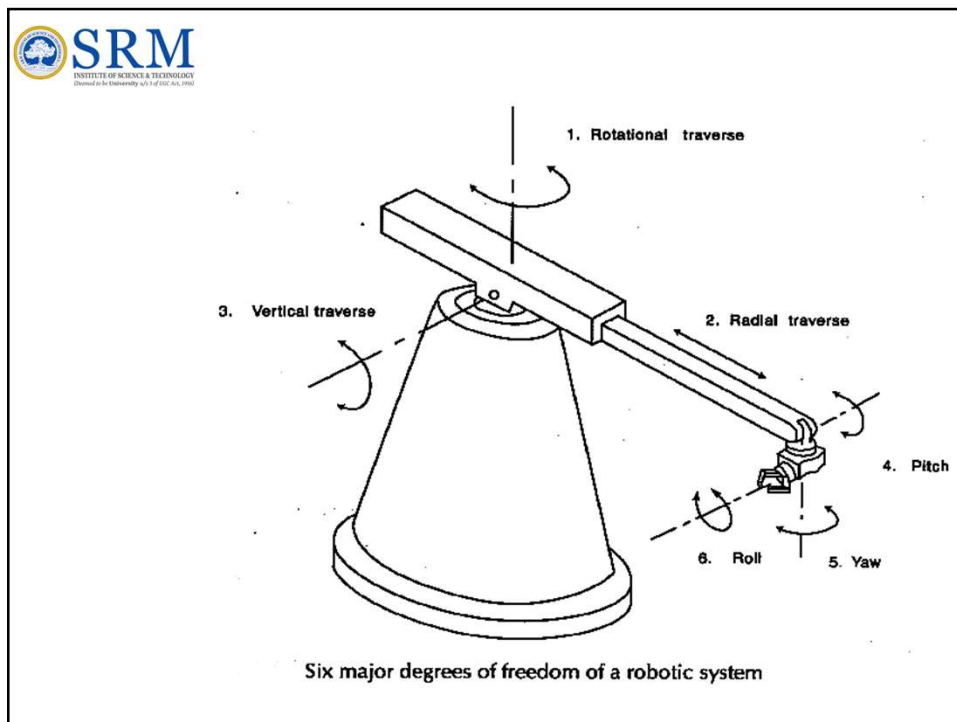


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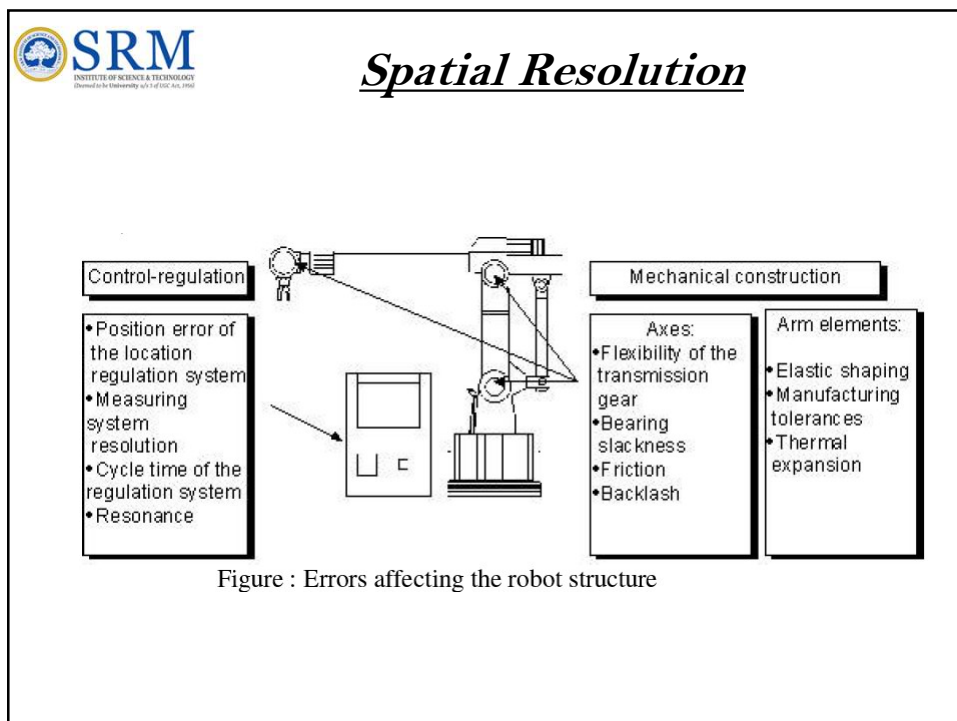
Six basic robot motions are ???

1. **Radial traverse:** Involve the extension and retraction (in or out movement) of the arm relative to the base
2. **Vertical traverse:** Provide up-and-down motion of the arm
3. **Rotational traverse:** Rotation of the arm about vertical axis such as left-and-right swivel of the robot arm about a base
4. **Wrist Pitch/Bend:** Provide up-and-down rotation to the wrist
5. **Wrist Yaw:** Involve right-and-left rotation of the wrist
6. **Wrist Roll/Swivel:** Is the rotation of the wrist about the arm axis

70



71



72



Spatial Resolution

- Defined as smallest increment of movement into which the robot can divide its work volume
- Depends on two factors: system's control resolution and the robot's mechanical inaccuracies
- Control resolution is determined by robot's position control system and its feedback measurement system
- Ability to divide total range of movement for the particular joint into individual increments that can be addressed in the controller
- The increments are sometimes referred to as “addressable points”

73



SPATIAL RESOLUTION

- Joint range depends on the bit storage capacity in the control memory
- Number of increments for a axis is given by
- $\text{Number of Increments} = 2^n$
- n = the number of bits in the control memory
- Have a control resolution for each joint in case of several DOF
- Total control resolution depend on the wrist motions as well as the body and arm motions

74



SPATIAL RESOLUTION

- Mechanical inaccuracies come from elastic deflection in the structure elements, gear backlash, stretching of pulley cords, leakage of hydraulic fluids and other imperfections in the mechanical system
- Also affected by load being handled, the speed of arm moving, condition of maintenance of robot

75



Accuracy

- Ability to position its wrist end at a desired target point within the work volume
- Depends on spatial resolution means how closely the robot can define the control increments
- One half of the control resolution by ignoring the mechanical inaccuracies

76

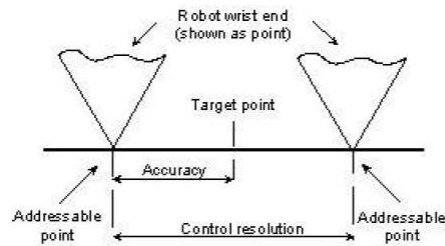


Figure : Diagram of accuracy in two dimensions frame, without mechanical inaccuracy consideration

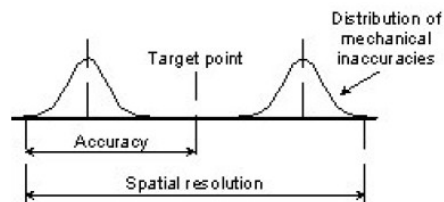


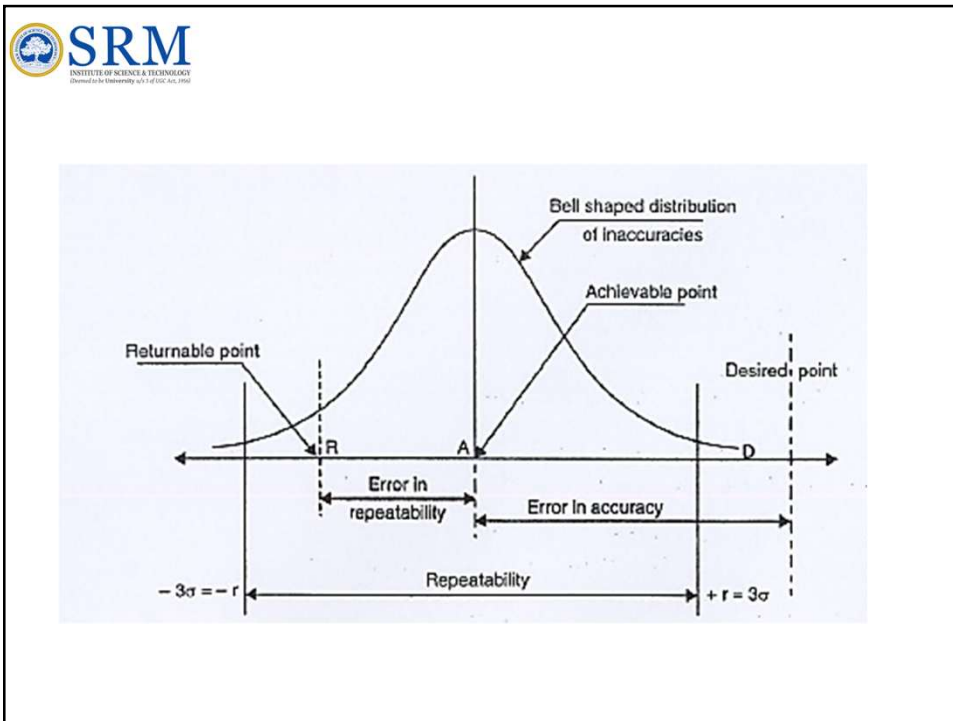
Figure: Diagram of accuracy and spatial resolution in which mechanical inaccuracies are represented by a statistical distribution

77

Repeatability

- Ability to position its wrist at a point in space that had been taught
- Accuracy relates to its capacity to be programmed to achieve a given target point
- Programmed point and target point may be different due to limitations of resolution
- Repeatability refers to ability to return to the programmed point when commanded to do so

78



79

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Robot drive systems

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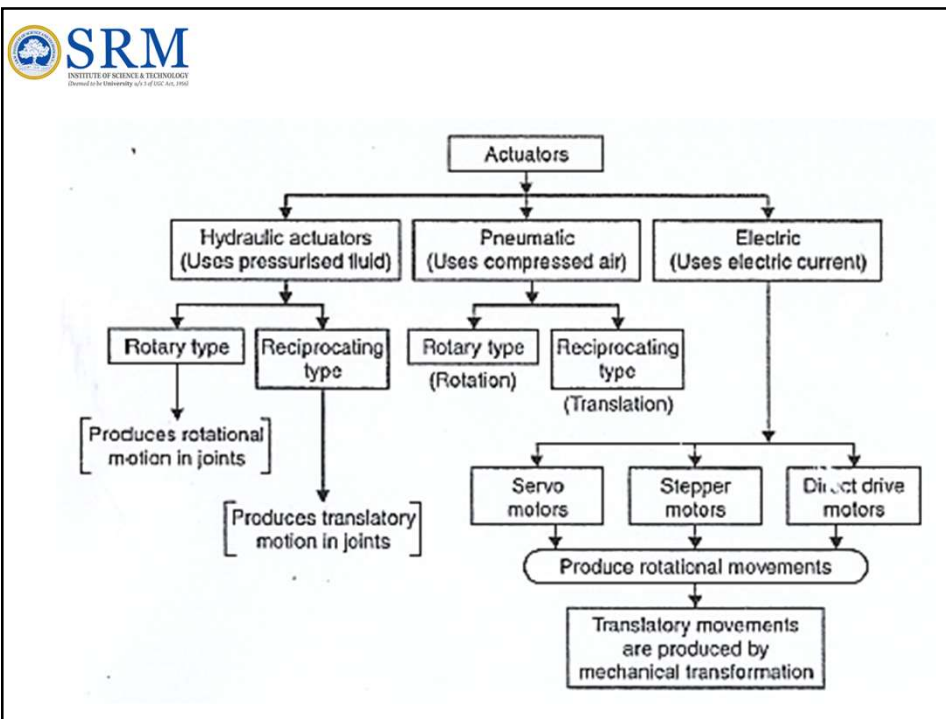
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Drive Systems

- Capacity to move robot's body, arm and wrist
- Determine speed of the arm movements, strength of the robot & dynamic performance
- Type of applications that the robot can accomplish
- Powered by three types of drive systems:
 - Hydraulic
 - Pneumatic
 - Electric

81



82

Hydraulic Drive

1. Linear piston or rotary vane types.
2. High power-to-weight ratio suitable for high loads at moderate speeds.
3. Easily controllable due to high stiffness of hydraulic system.
4. Require high energy storage systems such as pumps and accumulators.
5. Susceptible to leakage, which may reduce efficiency.
6. Require filtration.
7. Air entrapment and cavitation may cause difficulties.
8. Servo valves controlling the fluid flow must be placed close to the actuators to increase the stiffness of the control system.
9. Suitable for harsh environments caused by dust, dirt, and moisture.

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83

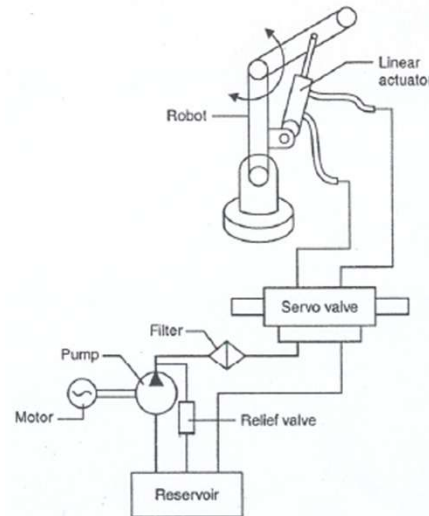
83

1. Hydraulic Drive

- Associated with large robot
- Provide greater speed & strength
- Add floor space
- Leakage of oil
- Provide either rotational or linear motions
- Applications such as:
 - Spray coating robot
 - Heavy part loading robot
 - Material handling robot
 - Translatory motions in cartesian robot
 - Gripper mechanism

84

1. Hydraulic Drive..



85

Pneumatic actuators

1. Used in inexpensive manipulators with low load-carrying capacity.
2. Usually require mechanical stops for accurate positioning in non-servo control.
3. Single- and double-acting piston or rotary vane types are commonly used.
4. Inherently lightweight at moderate operating pressures.
5. Limited by low efficiency at reduced loads and low stiffness due to compressibility.

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86

86



2. Pneumatic Drive

- Reserved for smaller robot
- Limited to “pick-and-place” operations with fast cycles
- Drift under load as air is compressible
- Provide either rotational or linear motions
- Simple and low cost components
- Used to open and close gripper

87



Electric motors

1. Provide a variety of power systems.
2. Easier to control.
3. May not be as responsive as hydraulic systems and stiffer than pneumatic systems with low pressure.
4. Normally require speed reducers, which introduce inefficiency and error in operation.
5. Variety of types such as permanent magnet DC motors, printed circuit motors, stepping motors, and direct drives.

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88

88



3. Electric Drive

- Rotor, stator, brush and commutator assembly
- Rotor has got windings of armature and stator has got magnets
- The brush and the commutator assembly switch the current in armature windings
- The most commonly used are DC servomotors, AC servomotors and stepper motors

89



Robot Controls with Hydraulic, Pneumatic, and Electric Systems

Hydraulic	Pneumatic	Electric
Pressure control	Pressure regulators	Current limiters
Motion control	Pneumatic motion controls	Mechanical motion controllers
Flow control	Flow restricters	Current limiters
Locking control	Pneumatic locking controls	Mechanical brake or self locking worm gear

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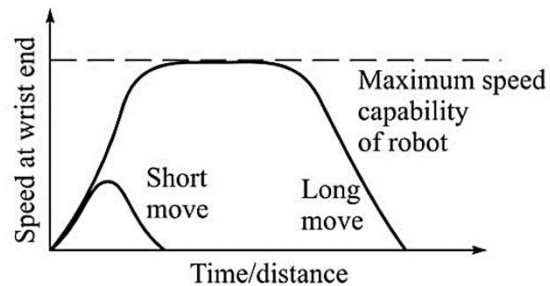
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90

SRM Institute of Science & Technology Speed of motion and Load carrying capacity

Speed depends on the following factors

- The accuracy with which the end effector must be positioned
- The weight of the object being manipulated
- The distance to be moved



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91

91

SRM Institute of Science & Technology Speed of motion and Load carrying capacity

Can you tell me how much load you can carry??

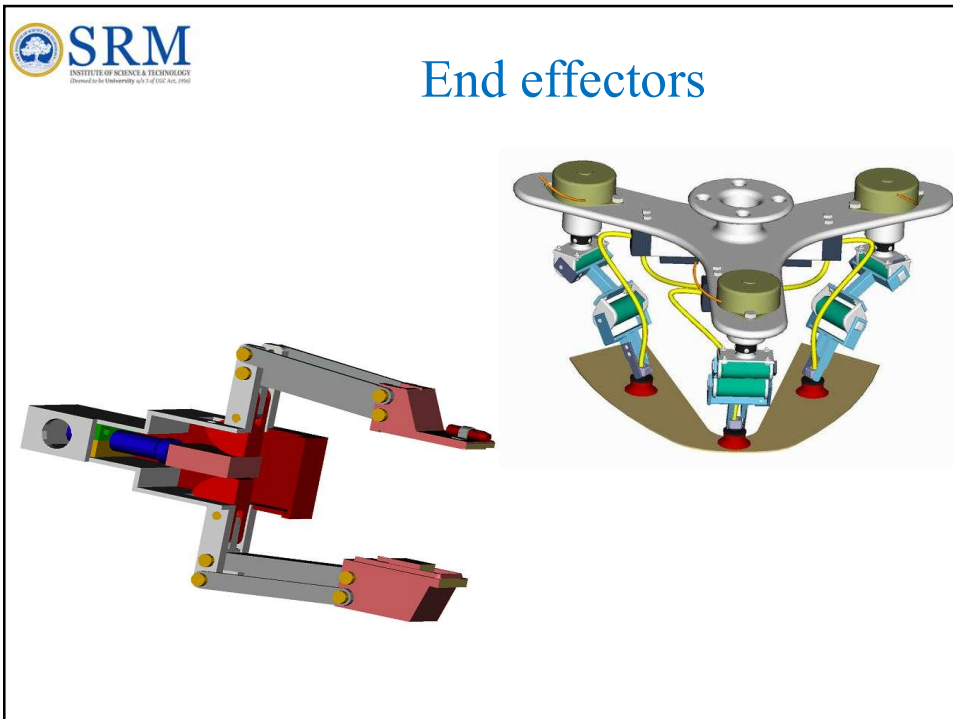


The load carrying capacity should be specified under the condition that the robots arm is in its weakest position.

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92

92



93

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Classification

Two categories of end effectors are

- Grippers
- Tools

According to the types of grasping, the end effectors are grouped in to

- Mechanical fingers
- Special tools
- Universal fingers

94



Tools are used in applications where the robot must perform some processing operation on the part. Examples of the tools are:

- Spot welding gun.
- Arc welding tool.
- Spray painting gun.
- Rotating spindle for drilling, grinding, and so forth.
- Assembly tool (e.g. automatic screw driver)
- Heating torch.

95



Types of grippers

- Mechanical grippers
- Hooking or lifting grippers
- Grippers for scooping or ladling powders or molten metals or plastics
- Vacuum cups (Negative Pressure)
- Magnetic grippers
- Adhesive
- Electrostatic grippers

96



Mechanical grippers

- It uses mechanical fingers actuated by a mechanism to grip an object.
- Various mechanisms used in mechanical grippers are
 - Linkage
 - Gear and rack
 - Cam
 - Screw
 - Cable and pulley, etc.,

The grippers can actuate the opening or closing of the fingers by

- Pivoting or swinging movement
- Linear or translational movement

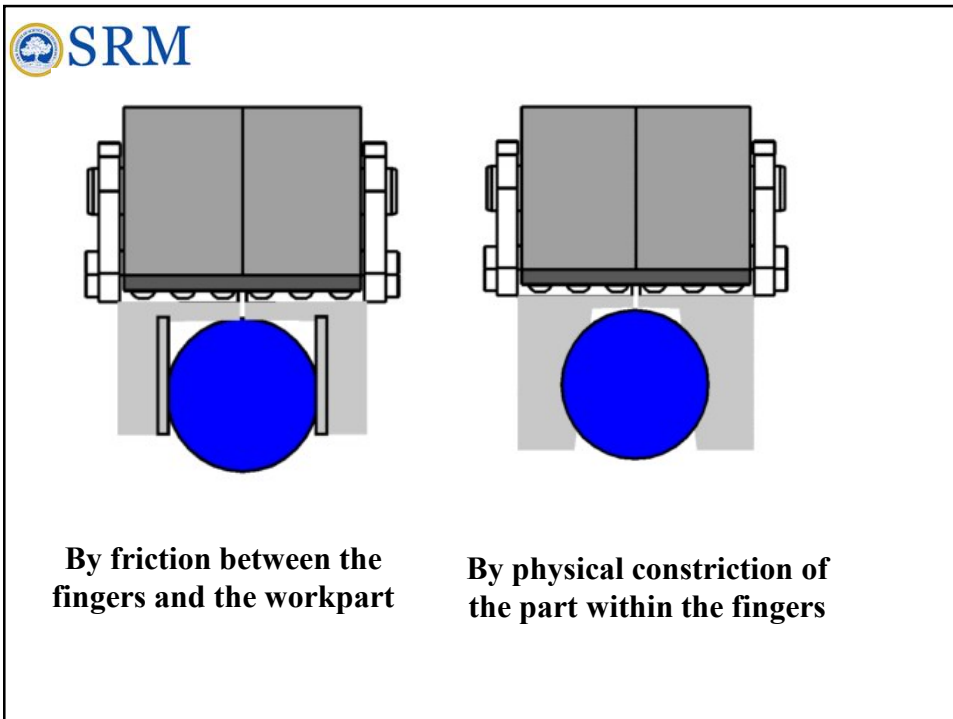
97



Mechanical grippers

- The function of the gripper mechanism is to translate some form of power input into the grasping action of the fingers against the part.
- There are two ways of constraining the part in the gripper.
 1. **By physical constriction of the part within the fingers** – this is usually accomplished by the contacting surfaces of the fingers to be in the approximate shape of the work geometry
 2. **By friction between the fingers and the workpart** – the fingers must apply a force that is sufficient for friction to retain the part against gravity, acceleration, and any other force that might arise during the holding portion of the work cycle.

98



99

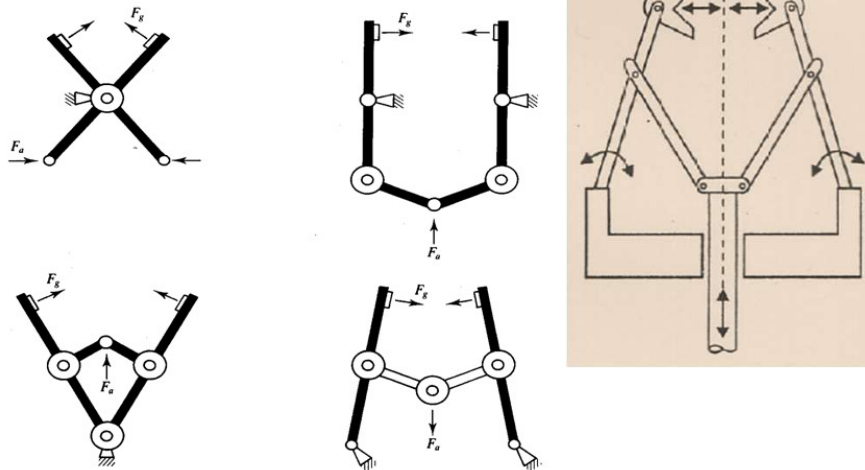
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- The part might slip out of the gripper if the applied force is insufficient.
- To resist this slippage, the gripper must be designed to exert a force that depends on the weight of the object, the co-efficient of friction between the part surface and the finger, the acceleration of the part and the orientation between the direction of motion during acceleration and the direction of fingers.

100

Types of mechanical actuation - Pivoting or swinging movement

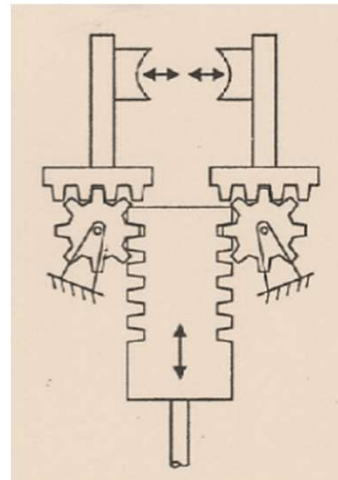
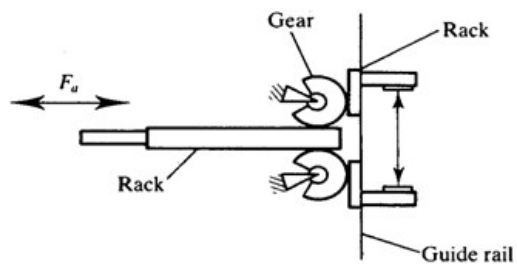
Linkage actuation:



101

Rack and Pinion actuation:

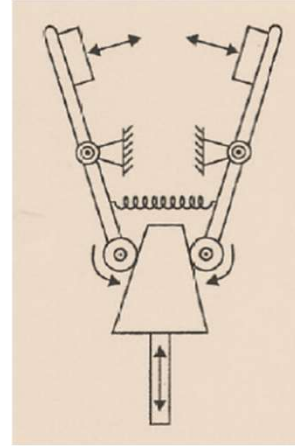
- A rack actuated by a hydraulic / pneumatic cylinder operates two pivoted pinions.
- The fingers attached to



102

CAM actuation:

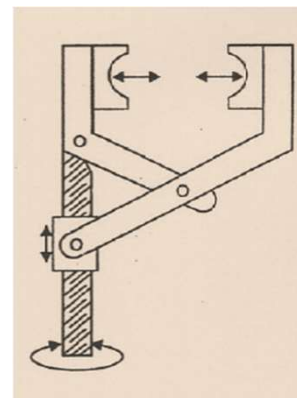
- A cam attached to a cylinder moves to and fro.
- Spring loaded followers moves on the cam profile.
- The finger pads attached to the pivoted lever connected to the followers open and close to release and grab the object



103

Screw actuation:

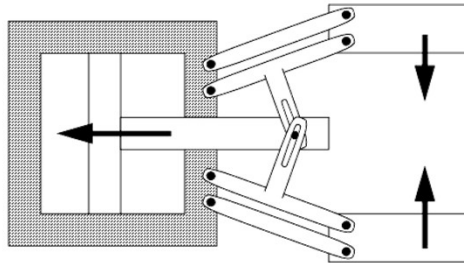
- A threaded block engages with a screw moves forward and backward, the fingers attached to levers connected to the threaded block by a hinged joint, open and close providing releasing and gripping action.
- The screw is rotated by a motor through reduction box.



104

Types of mechanical actuation – translational movement

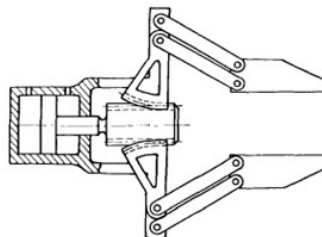
Parallel finger actuator - as the cylinder is actuated, the fingers move together and apart in parallel.



105

Gear and Rack Grippers

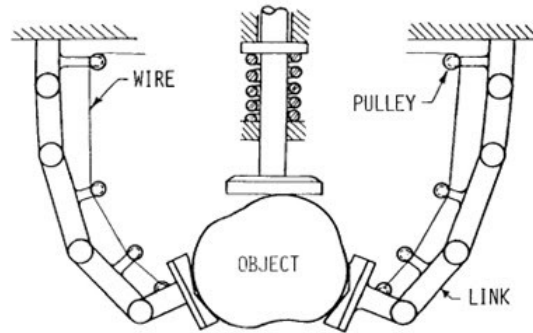
movement of input due to gear motion which makes connecting links to go in motion to make gripping action at the output link.



<http://sra.vjti.info/knowledgebase/posts-on-a-roll-1/gripper-mechanisms>

106

Three Fingered Grippers

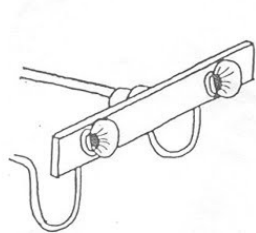


<http://sra.vjti.info/knowledgebase/posts-on-a-roll-1/gripper-mechanisms>

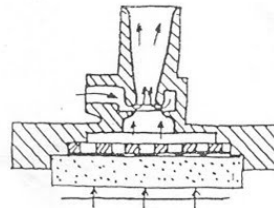
107

Vacuum Grippers [link](#)

for non-ferrous components with flat and smooth surfaces, grippers can be built using standard vacuum cups or pads made of rubber-like materials. Not suitable for components with curved surfaces or with holes.



Vacuum gripper with suction cups

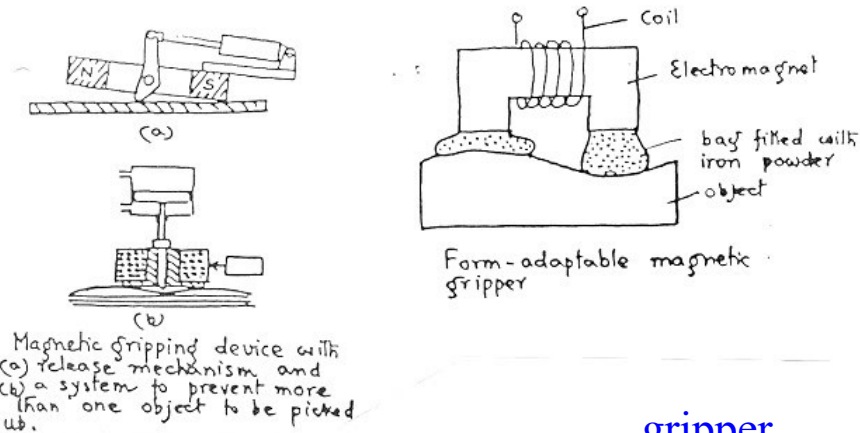


Venturi vacuum gripper for flat plates

<http://sra.vjti.info/knowledgebase/posts-on-a-roll-1/gripper-mechanisms>

108

Magnetic Gripper



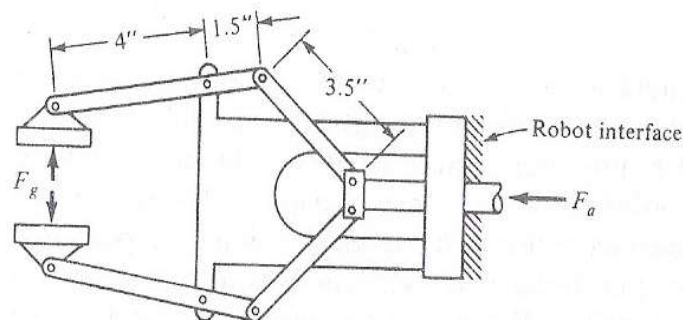
gripper

<http://sra.vjti.info/knowledgebase/posts-on-a-roll-1/gripper-mechanisms>

109

GRIPPER FORCE ANALYSIS

The fig. shows the linkage mechanism and dimensions of a gripper used to handle a workpart for a machining operation. Suppose it has been determined that the gripper force is to be 25 N. What is required to compute the actuating force to deliver this force of 25 N.



110

110



Due to symmetry of the gripper consider only one half of the mechanism.

Moments :-

$$25(4\cos 15^\circ) = F\sin 45^\circ(1.5\cos 15^\circ) + F\cos 45^\circ(1.5\sin 15^\circ)$$

$$96.6 = F(1.0246 + 0.2745) = 1.2991F$$

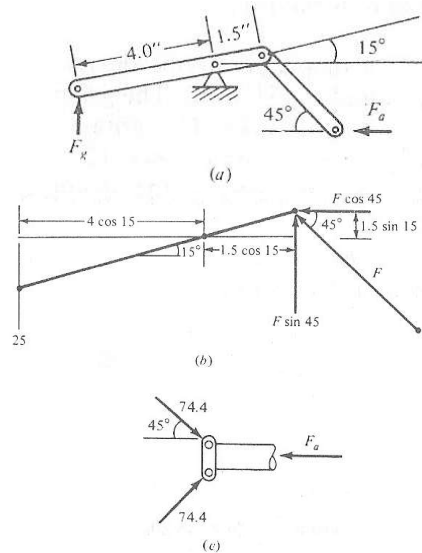
$$F = 74.4 \text{ N}$$

The actuation force applied to the plunger to deliver this force of 74.4 N to each finger.

So,

$$F_a = 2 \times 74.4 \times \cos 45^\circ$$

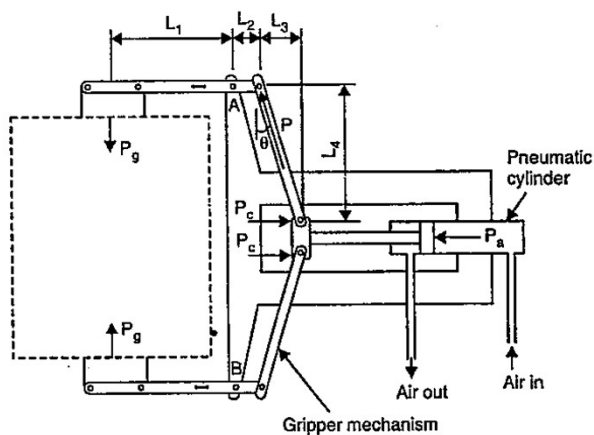
$$F_a = 105.2 \text{ N}$$



111



GRIPPER FORCE ANALYSIS



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112

112



The friction force is given by,

$$f = \mu NP_g,$$

μ = coefficient of friction,

N = the number of fingers.

$$F_d = \text{design force} = n.W$$

factor of safety, n .

$$n.W = \mu NP_g$$

$$P_g = \frac{n.W}{\mu N} = \frac{n.m.g}{\mu N}$$

If the gripper is accelerating or decelerating by ' a '

$$P_g = \frac{n.m}{\mu N} (g \pm a)$$

113



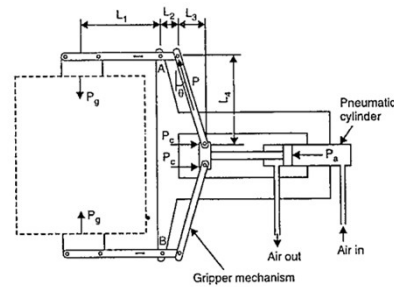
$$P_g = \frac{n.m}{\mu N} (g \pm a)$$

$$P_g = \frac{n.(mg)}{\mu N} (1 \pm K_f)$$

The factor $K_f = \frac{a}{g}$ and weight of the object, $W = (mg)$

Accelerating motion	Factor $(1 \pm K_f)$
• Accelerating down	3
• Accelerating up	1
• Accelerating in horizontal direction	2

114



taking moment about the point A,

$$P_g \cdot L_1 - (P \cos \theta) \cdot L_2 = 0$$

$$P = \frac{P_g \cdot L_1}{(L_2 \cdot \cos \theta)}$$

where $\theta = \tan^{-1} \left(\frac{L_3}{L_4} \right)$ and P = linkage force in Newton

115

$$P_c = P \cdot \sin \theta$$

Substituting equation

$$P_c = \frac{P_g \cdot L_1}{L_2} \cdot \tan \theta.$$

$$P_a = 2P_c$$

$$P_a = \frac{2nW}{\mu N} \left(\frac{L_1}{L_2} \right) (1 \pm K_f) \cdot \tan \theta$$

$$\tan \theta = \frac{L_3}{L_4}$$

$$P_a = \frac{2nW}{\mu N} \left(\frac{L_1 \cdot L_3}{L_2 L_4} \right) (1 \pm K_f)$$

116



Power requirements:

If the diameter of the piston of the actuator is d_p

Area of the piston, $A_p = \frac{\pi}{4} \cdot d_p^2$ in m^2

The pressure of air needed $= \frac{P_a}{A_p} = p$

$$p = \frac{4}{\pi} \cdot \frac{P_a}{d_p^2}$$

The power required to produce this pressure, $p(N/m^2)$

$$P_R = \frac{p \cdot Q}{1000} \text{ kW}$$

where Q is the discharge (volume flow rate) of air in m^3/sec .

117



A Cartesian robot has a horizontal reach of 450 mm and a horizontal stroke of 250 mm. What is the maximum limit within which the object placed is not reachable?

H = maximum horizontal reach = 450 mm.

S = stroke length = 250 mm.

L = non-reachable maximum limit = ?

$$L = (H - S)$$

$$= (450 - 250) = 200.$$

Non-reachable maximum limit = 200 mm.

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118

118



One of the links of a robot has a telescoping arm with a stroke of 512 mm. The control memory of the robot has 8-bit storage capacity for this axis. Determine the control resolution for the same.

$S = \text{stroke length} = 512 \text{ mm.}$

$K = \text{storage capacity} = 8 \text{ bit memory.}$

From the equation (2.2),

Total control resolution,

$$R_t = \frac{S}{(2)^k} = \frac{512}{(2)^8} = 2 \text{ mm.}$$

Ans. Total control resolution = 2 mm.

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119

119



CONSIDERATIONS IN GRIPPER SELECTION AND DESIGN

The important factors that determine the required grasping force are:

- The weight of the object. Consideration of whether the part can be grasped consistently about its center of mass. If not, an analysis of the possible moments from off-center grasping should be considered.
- The speed and acceleration with which the robot arm moves (acceleration and deceleration forces), and the orientational relationship between the direction of movement and the position of the fingers on the object (whether the movement is parallel or perpendicular to the finger surface contacting the part)
- Whether physical constriction or friction is used to hold the part.
- Coefficient of friction between the object and the gripping fingers.

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120

120

2.20 GRIPPER DESIGN CONSIDERATION

<i>Part Specification</i>	<i>Performance Specification</i>	<i>Source Specification</i>	<i>Position Specification</i>	<i>Environmental Specification</i>	<i>Material Specification</i>
- Weight of the part - Size and shape of the component - Tolerance on the part size. - Change of shape and size during processing. - Surface finish of the part. - Care for delicacy of the part to be handled.	- Co-efficient of friction between part and object. - Speed and acceleration during motion. - Accuracy and repeatability of the robot. - Interchangeability of fingers. - Memory capacity of the controller. - Control resolution of the robot. - Lead time and cycle time.	- Pneumatic: air pressure and discharge, and cylinder size. - Electrical: The power rating and specification of actuator. - Hydraulic: The oil pressure, volume flow rate and the power pack specification. - Mechanical: Linkage design and transmission design specification.	- Holding methods. - Physical difficulty - Length of fingers - The object and tool orientation. - Product design changes. - Spare parts specifications.	- Heat and temperature of object and atmosphere. - Humidity and moisture. - Dirty and safety. - Hazardness of chemicals used.	- Strength and rigidity. - Durability and fatigue factors. - Friction properties. - Factor of safety. - Design standards. - Compatibility with the environment.

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121

121

Gripper design considerations

<i>Part Specification</i>	<i>Performance Specification</i>	<i>Source Specification</i>
- Weight of the part - Size and shape of the component - Tolerance on the part size. - Change of shape and size during processing. - Surface finish of the part. - Care for delicacy of the part to be handled.	- Co-efficient of friction between part and object. - Speed and acceleration during motion. - Accuracy and repeatability of the robot. - Interchangeability of fingers. - Memory capacity of the controller. - Control resolution of the robot. - Lead time and cycle time.	- Pneumatic: air pressure and discharge, and cylinder size. - Electrical: The power rating and specification of actuator. - Hydraulic: The oil pressure, volume flow rate and the power pack specification. - Mechanical: Linkage design and transmission design specification.

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122

122

Gripper design considerations

<i>Position Specification</i>	<i>Environmental Specification</i>	<i>Material Specification</i>
• Holding methods. • Physical difficulty • Length of fingers • The object and tool orientation. • Product design changes. • Spare parts specifications.	• Heat and temperature of object and atmosphere. • Humidity and moisture. • Dirty and safety. • Hazardness of chemicals used.	• Strength and rigidity. • Durability and fatigue factors. • Friction properties. • Factor of safety. • Design standards. • Compatibility with the environment.

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123

123

Selection consideration of gripper

<i>Actuation Selection</i>	<i>Drive Selection</i>	<i>Protection Selection</i>	<i>Process Selection</i>
• Mechanical or friction gripping methods. • Pad shape selection. • Vacuum actuation • Magnetic grasping. • Adhesive gripping. • Expandable bladder type actuation.	• Pneumatic drive systems. • Hydraulic drive System for heavy duty operation. • Electrical drive for light duty application. • Speed reduction of the mechanical transmission.	• Heat shield for sensors and actuators. • Forced cooling by air or water cooling to take away the heat. • Selection of heat resistance materials for fingers and components of gripper. • Shield from hazardous chemicals.	• Accurate processing methods for fingers. • Leak prevention for pneumatic/hydraulic actuators. • Interchangeability standards for the fingers. • Shape compatible processing methods for fingers. • Ease of assembly of fingers and linkages.

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124

124



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Checklist of factors in the selection and design of grippers

Factor	Consideration
Part to be handled	Weight and size Shape Changes in shape during processing Tolerances on the part size Surface condition, protection of delicate surfaces
Actuation method	Mechanical grasping Vacuum cup Magnet Other methods (adhesives, scoops, etc.)

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125

125



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Checklist of factors in the selection and design of grippers

Factor	Consideration
Power and signal transmission	Pneumatic Electrical Hydraulic Mechanical
Gripper force (mechanical gripper)	Weight of the object Method of holding (physical constriction or friction) Coefficient of friction between fingers and object Speed and acceleration during motion cycle
Positioning problems	Length of finger's Inherent accuracy and repeatability of robot Tolerances on the part size

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126

126



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Checklist of factors in the selection and design of grippers

Factor	Consideration
Service conditions	Number of actuations during lifetime of gripper Replaceability of wear components (fingers) Maintenance and serviceability
Operating environment	Heat and temperature Humidity, moisture, dirt, chemicals
Temperature protection	Heat shields Long fingers Forced cooling (compressed air, water cooling, etc.) Use of heat-resistant materials

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127

127



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Checklist of factors in the selection and design of grippers

Factor	Consideration
Fabrication materials	Strength, rigidity, durability Fatigue strength Cost and ease of fabrication Friction properties for finger surfaces Compatibility with operating environment
Other considerations	Use of interchangeable fingers Design standards Mounting connections and interfacing with robot Risk of product design changes and their effect on the gripper design Lead time for design and fabrication Spare parts, maintenance, and service

10/3/2025
128

128

